

Integration Architecture

Integration Summary

The platform integrates through:

- Microsoft Entra ID for identity.
- Azure Service Bus for internal workflow events.
- SMTP/email for notifications.
- Payroll providers through the payroll broker.
- Azure OpenAI for AI analytics and semantic reasoning.
- Azure Blob Storage for files and model/certificate artifacts.
- Future HRIS, payroll, BI, and notification connectors.

Current Integration Points

Integration	Direction	Current implementation
Microsoft Entra ID	Inbound identity	MSAL frontend and JWT validation backend.
Azure SQL	Read/write	Backend, workers, analytics job, payroll broker.
Azure Service Bus	Internal eventing	Backend, integrity worker, auxiliary worker, payroll broker.
Email/SMTP	Outbound notification	Auxiliary handlers.
Azure Blob Storage	Read/write files	Fraud models, certificates, templates, exports.
Azure OpenAI	Outbound AI calls	Description checks, explanations, analytics agents.
Gusto	Payroll provider	Payroll broker provider and webhooks.
Rippling	Payroll provider	Payroll broker provider structure and webhooks.
Workday pattern	Payroll/webhook pattern	Backend webhooks and payout/proxy references.
Log Analytics	Observability	KQL assets and GitHub workflows.

Identity Integration

Microsoft Entra ID is the identity provider. The platform expects:

- Frontend app registration.
- API app registration.
- API scope such as `api://<CLIENT_ID>/access_as_user`.
- App roles for administrators.
- Tenant registration in `Tenants`.
- User roster import into `Users`.

Future identity integrations:

- B2B guest onboarding automation.
- SCIM user provisioning.
- Group-to-role synchronization.
- Conditional Access alignment.

HRIS Integration Direction

Current user and manager data is stored in **Users**, seeded/imported through scripts. Future HRIS integration should synchronize:

- Employee ID.
- Name.
- Work email.
- Department/title.
- Manager relationship.
- Employment status.
- Location/country.
- Cost center.

Potential providers:

- Workday.
- Oracle HCM.
- SAP SuccessFactors.
- ADP.
- Microsoft Graph / Entra ID.

Recommended pattern:

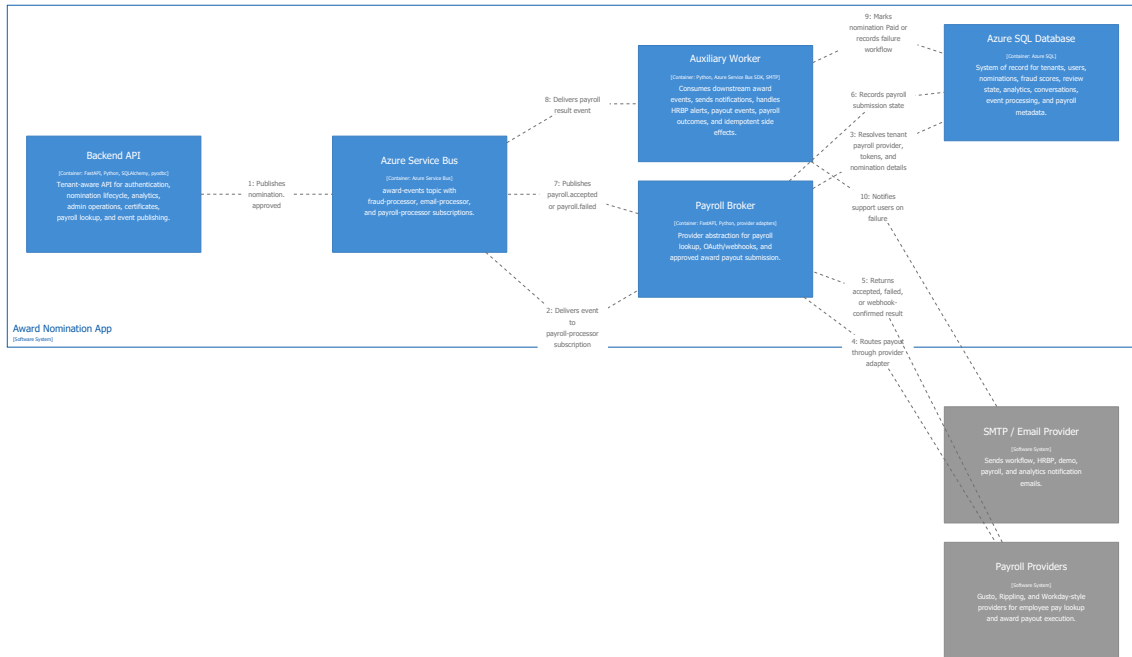
- Build an ingestion adapter per provider.
- Normalize into a tenant-scoped employee model.
- Stage incoming data before applying changes.
- Detect manager-cycle and missing-manager errors.
- Record sync run metadata and row-level errors.

Payroll Integration

The payroll broker abstracts provider-specific payout and lookup behavior.

Current pattern:

Structured source: `../diagrams/structurizr/workspace.dsl`, view **PayrollIntegrationFlow**.



Dynamic View: Award Nomination App
Approved nomination payout flow through the payroll broker and provider registry.
Tuesday, July 7, 2026 at 3:48 PM Pacific Daylight Time

Payroll integration flow

Provider responsibilities:

- OAuth onboarding.
- Token refresh.
- Employee lookup.
- Off-cycle payroll submission.
- Webhook verification.
- Callback handling.

Recommended provider contract:

- `get_employee_profile(upn, tenant_id)`
- `get_employee_pay(upn, year, month)`
- `submit_award_payout(nomination)`
- `verify_webhook(request)`
- `handle_webhook(payload)`

Email Integration

Auxiliary worker sends emails for:

- Manager approval request.
- Nomination approved/rejected outcome.
- HRBP review request.
- HRBP review outcome.
- Description rejection.
- HRBP SLA breach.

- Demo access request.
- Payroll failure.
- Free-form analytics notification.

Email templates are tenant/language-aware through [EmailTemplates](#).

Recommended future enhancements:

- Provider abstraction for SMTP, Microsoft Graph Mail, SendGrid, or Azure Communication Services.
- Delivery status tracking.
- Bounce/failure tracking.
- Tenant-specific sender domain verification.

Analytics Export Integration

Current export paths include:

- Integrity finding Excel export.
- Ask/investigate export payloads.
- MCP analytics export service.

Future export targets:

- Power BI datasets.
- Azure Data Lake.
- Snowflake/Databricks.
- Scheduled executive PDF/email reports.
- SIEM event export.

Webhook Integration

Webhook patterns exist for:

- Workday payment confirmation.
- Gusto payroll submitted callbacks.
- Rippling payroll run completed callbacks.
- Internal model refresh callback.
- Internal HRBP SLA callback.

Webhook requirements:

- Validate signature or shared secret.
- Enforce idempotency by provider reference.
- Persist raw callback metadata when needed for audit.
- Avoid logging secrets or full sensitive payloads.

Event Integration Contract

Service Bus event envelope:

```
{
  "event_type": "nomination.approved",
  "nomination_id": 42,
  "timestamp": "2026-07-07T00:00:00Z"
}
```

Optional **extra** payload can carry event-specific data such as:

- Reviewer ID.
- HRBP rejection reason.
- Payment reference.
- Notification fields.

Event metadata:

- **event_type** is also set as a Service Bus application property for broker-side SQL filters.
- Trace headers are propagated through application properties.

Future Integration Roadmap

- HRIS roster sync.
- SCIM provisioning.
- Microsoft Teams notifications.
- Power BI semantic model.
- Webhook subscription API for tenants.
- Payroll provider certification checklist.
- Tenant-specific outbound webhook signing.
- Data export API for audit and compliance.